

Supplementary Information

Permitless Concealed Carry Adoption and Firearm Mortality in U.S. States, 1999–2024

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June 2026

Supplementary Note 1. Evidence hierarchy

The main manuscript interprets results through a hierarchy rather than treating every diagnostic row as an independent hypothesis test. Firearm suicide is the primary outcome. Total suicide, non-firearm suicide, firearm homicide, and total firearm deaths are secondary outcome-family checks. Firearm-law covariates, contextual covariate layers, staggered-adoption diagnostics, Arkansas recoding, leave-one-out influence, and placebo timing are sensitivity checks. Rurality, gun ownership, baseline-risk heterogeneity, and policy-feature interactions are exploratory. Cohort ATT rows, event-time rows, full legal source tables, and full robustness grids are appendix diagnostics.

Tier	Items	Interpretation
Primary outcome	Firearm suicide	Central mortality claim.
Secondary outcomes	Total suicide; non-firearm suicide; firearm homicide; total firearm deaths	Specificity and outcome-family interpretation.
Sensitivity checks	Firearm-law covariates; health-access and overdose covariates; Phase 3B2 covariate layers; staggered-adoption diagnostics; Arkansas recoding; leave-one-out; placebo timing	Tests stability and boundaries.
Exploratory checks	Rurality, gun-ownership, baseline-risk, and policy-feature heterogeneity	Hypothesis-generating only.
Appendix diagnostics	Cohort ATT rows; event-time rows; legal source table; full robustness grids	Transparency and audit trail.

Supplementary Note 2. Legal audit and policy-feature coding

The legal audit records one row per state and distinguishes clean source-verified within-panel adopters from verified non-adopters, partial cases, ambiguous cases, and baseline-permitless states. Partial, ambiguous, and baseline-permitless rows are documented so that they are not silently absorbed into the binary treatment definition.

Audit status	State count
Source-verified current adopter	26
Verified non-adopter	21
Ambiguous reviewed	1
Partial	1
Baseline permitless	1

Among the 26 clean source-verified adopter rows, 21 had a training requirement removed, 25 removed the carry-permit background-check screen, and 12 removed a permit-specific misdemeanor-violence screen. These policy-feature fields describe the carry-permit process and are treated as exploratory.

Mechanism field/value	State count
Training requirement removed: yes	21
Training requirement removed: no prior training requirement	5
Carry-permit background-check screen removed	25
Carry-permit screen removed while purchase permit retained	1
Permit-specific misdemeanor screen removed	12
Eligibility standard retained but no pre-carry check	7
Other misdemeanor-screen coding categories	7

The full state-level audit below reports the adoption year, audit status, whether the state enters the primary processed treatment map, the coding note, and source URLs used for legal verification. This table is included to make Arkansas, Mississippi, and Vermont visible rather than hiding the edge cases inside a binary treatment variable.

Table 1: State-by-state permitless concealed-carry treatment coding audit.

State	Year	Audit status	Primary map	Coding note	Source URL(s)
Alabama	2023	clean adopter	Yes	Clean source-verified within-panel adopter.	https://legiscan.com/AL/text/HB272/id/2544995 ; https://www.handgunlaw.us/documents/Permitless'Ca
Alaska	2003	clean adopter	Yes	Clean source-verified within-panel adopter.	https://dps.alaska.gov/Statewide/R-1/PermitsLicensing/Firearm-FAQ ; https://www.handgunlaw.us/documents/Permitless'Ca
Arizona	2010	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.azleg.gov/legtext/49leg/2r/bills/sb1108s ; https://www.handgunlaw.us/documents/Permitless'Ca
Arkansas	–	ambiguous reviewed	No	Excluded from primary clean-adoption map; 2021 and 2023 codings tested as sensitivities.	https://arkleg.state.ar.us/Bills/Detail?ddBienniumSess ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/
California	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ; https://oag.ca.gov/firearms/ccw
Colorado	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ; https://cbi.colorado.gov/firearms/concealed-handgun-permit-chp-reciprocity
Connecticut	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ; https://portal.ct.gov/despp/division-of-state-police/special-licensing-and-firearms/pistol-permits
Delaware	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ; https://attorneygeneral.delaware.gov/criminal/concealed-carry-deadly-weapons-ccdwl
Florida	2023	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.flgov.com/eog/news/press/2023/governor-ron-desantis-signs-hb-543-constitutional-carry ; https://www.handgunlaw.us/documents/Permitless'Ca
Georgia	2022	clean adopter	Yes	Clean source-verified within-panel adopter.	https://gov.georgia.gov/press-releases/2022-04-13/gov-kemp-signs-georgia-constitutional-carry-act-law ; https://www.handgunlaw.us/documents/Permitless'Ca

State	Year	Audit status	Primary map	Coding note	Source URL(s)
Hawaii	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ; https://www.honolulupd.org/police-services/firearms/
Idaho	2016	clean adopter	Yes	Clean source-verified within-panel adopter.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.handgunlaw.us/documents/Permitless'Ca
Illinois	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ; https://isp.illinois.gov/Foid/Ccl https://events.in.gov/event/gov'holcomb'issues'stateme https://www.in.gov/isp/files/Permitless-Carry-Website-Messaging.pdf
Indiana	2022	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.legis.iowa.gov/legislation/billTracking/bil https://www.handgunlaw.us/documents/Permitless'Ca
Iowa	2021	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.kslegislature.gov/li/2016/b2015'16/measu https://www.handgunlaw.us/documents/Permitless'Ca
Kansas	2015	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.kentuckystatepolice.ky.gov/ccdw-home/ ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/
Kentucky	2019	clean adopter	Yes	Clean source-verified within-panel adopter.	https://legis.la.gov/Legis/BillInfo.aspx?i=245605 ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/
Louisiana	2024	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.maine.gov/dps/msp/licenses-permits/concealed-carry-maine ; https://legislature.maine.gov/legis/bills/display'ps.asp
Maine	2015	clean adopter	Yes	Clean source-verified within-panel adopter.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://mdsp.maryland.gov/Organization/Pages/Crim
Maryland	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ; https://www.mass.gov/how-to/apply-for-a-firearms-license
Massachusetts	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ; https://www.michigan.gov/msp/services/ccw
Michigan	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ; https://dps.mn.gov/divisions/bca/bca-divisions/administrative/Pages/Permit-to-Carry.aspx https://billstatus.ls.state.ms.us/documents/2016/html/0799/HB0786PS.htm ; https://www.handgunlaw.us/documents/Permitless'Ca
Minnesota	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://www.senate.mo.gov/BillTracking/Bills/BillInfo https://www.handgunlaw.us/documents/Permitless'Ca
Mississippi	2015	partial	Yes	Retained in processed panel as 2015 adoption; legal audit marks the row partial.	https://legiscan.com/MT/bill/HB102/2021 ; https://www.handgunlaw.us/documents/Permitless'Ca
Missouri	2017	clean adopter	Yes	Clean source-verified within-panel adopter.	https://nebraskalegislature.gov/bills/view'bill.php?Doc https://statepatrol.nebraska.gov/services/concealed-handgun-permits/chp-frequently-asked-questions
Montana	2021	clean adopter	Yes	Clean source-verified within-panel adopter.	
Nebraska	2023	clean adopter	Yes	Clean source-verified within-panel adopter.	

State	Year	Audit status	Primary map	Coding note	Source URL(s)
Nevada	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ; https://rccd.nv.gov/FeesForms/Brady/CCW-Permit-Recognition/ ; https://legiscan.com/NH/votes/SB12/2017 ; https://www.handgunlaw.us/documents/Permitless'Ca
New Hampshire	2017	clean adopter	Yes	Clean source-verified within-panel adopter.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ;
New Jersey	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ; https://www.nj.gov/njsp/firearms/forms.shtml https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ; https://www.dps.nm.gov/concealed-carry/
New Mexico	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ;
New York	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ; https://gunsafety.ny.gov/pistol-permit-recertification https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ;
North Carolina	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ; https://ncdoj.gov/law-enforcement-training/criminal-justice/officer-certification-programs/concealed-weapon-reciprocity/
North Dakota	2017	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.nd.gov/ndhp/node/116 ; https://legiscan.com/ND/bill/HB1169/2017
Ohio	2022	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.legislature.ohio.gov/download?key=1896 ; https://www.handgunlaw.us/documents/Permitless'Ca
Oklahoma	2019	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.oklegislature.gov/cf/pdf/2019-20%20SUPPORT%20DOCUMENTS/BILLSUM/Hou
Oregon	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://www.handgunlaw.us/documents/Permitless'Ca https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/ ; https://www.oregon.gov/osp/programs/cjis/pages/oregon-concealed-handgun-license.aspx
Pennsylvania	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/state-laws/concealed-carry-in-pennsylvania/ ; https://giffords.org/lawcenter/state-laws/open-carry-in-pennsylvania/ ; https://www.pa.gov/en/agencies/psp/programs/firearms-in-pennsylvania.html
Rhode Island	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/ ; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/ ; https://riag.ri.gov/i-want/get-concealed-carry-permit
South Carolina	2024	clean adopter	Yes	Clean source-verified within-panel adopter.	https://sled.sc.gov/constitutional-carry-guidance ; https://legiscan.com/SC/supplement/H3594/id/47310
South Dakota	2019	clean adopter	Yes	Clean source-verified within-panel adopter.	https://sdlegislature.gov/Session/Bill/9374 ; https://www.handgunlaw.us/documents/Permitless'Ca
Tennessee	2021	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.wvlt.tv/2021/04/08/governor-lee-signs-permitless-carry-into-law/ ; https://www.handgunlaw.us/documents/Permitless'Ca

Table 2: **Firearm-suicide diagnostic spine.**

Diagnostic row	Estimate	Diag. p	Interpretation
Baseline unweighted fixed effects	1.263	< 0.001	Average state-level association
Population-weighted fixed effects	0.853	< 0.001	Resident-weighted association
Pre-COVID only	1.137	< 0.001	Drops post-2019 years
COVID-excluded	1.238	< 0.001	Drops 2020–2021
External firearm-law covariates	0.994	< 0.001	Policy-environment sensitivity
State-specific linear trends	0.253	0.069	Central attenuation caveat
Cohort ATT windows	0.591/0.553/0.627	–	2-, 3-, and 5-year windows
Not-yet-treated dynamic ATT	0.431	–	Positive post-adoption mean
Event-time fixed effects	0.617	0.000007	Positive post mean; diagnostic p is the minimum pre-adoption p

Estimates are deaths per 100,000 residents. TWFE rows include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors unless otherwise noted. Modern staggered-adoption rows are reported as diagnostics because event-time checks flag pre-adoption signal; the event-time p column reports the minimum pre-adoption p rather than a post-adoption effect test.

Supplementary Note 5. Firearm-law and contextual covariate sensitivity

External firearm-law covariates attenuated but did not erase the firearm-suicide estimate. Demographic-poverty covariates also preserved the firearm-suicide and total-suicide estimates. The short HRSA mental-health-provider window did not retain statistical significance and should be interpreted as a short-window sensitivity rather than a full-panel replacement. These rows are covariate-layer sensitivity checks, not uniformly superior causal models, because some time-varying covariates may lie downstream of broader state policy environments.

Firearm-suicide specification	Coef.	p	N
Baseline fixed effects	1.263	< 0.001	1300
External firearm-law covariates	0.994	< 0.001	1300
Health-access covariates	0.668	< 0.001	800
Overdose covariates	0.516	0.044	300
Demographic-poverty covariates	0.648	< 0.001	1000
Mental-health-provider access covariates	0.159	0.625	150
Full Phase 3B2 covariates	0.197	0.546	150

Supplementary Note 6. Change-score checks

Outcome	2-year diff.	3-year diff.	5-year diff.
Firearm suicide	0.641	0.603	0.674
Non-firearm suicide	0.219	0.151	0.033
Total suicide	0.860	0.754	0.707
Firearm homicide	0.016	0.081	0.112
Total firearm deaths	0.561	0.645	0.779

Supplementary Note 7. Staggered-adoption diagnostics

Outcome	ATT w2	ATT w3	ATT w5	Not-yet post	Interpretation
Firearm suicide	0.591	0.553	0.627	0.431	Positive post-adoption diagnostic with pretrend flag.
Non-firearm suicide	0.263	0.232	0.199	0.050	Positive post-adoption diagnostic.
Total suicide	0.854	0.785	0.827	0.481	Positive post-adoption diagnostic with pretrend flag.
Firearm homicide	0.085	-0.011	0.037	-0.030	No positive post-adoption diagnostic.
Total firearm deaths	0.391	0.454	0.557	0.360	Positive post-adoption diagnostic with pretrend flag.

Supplementary Note 8. Robustness and Arkansas sensitivity

Outcome	Baseline	P05 specs	Leave-one min	Leave-one max	Flag
Firearm suicide	1.263	5	1.135	1.356	Stable positive, trend-sensitive.
Non-firearm suicide	0.312	1	0.269	0.370	Positive but less stable.
Total suicide	1.576	6	1.420	1.727	Stable positive.
Firearm homicide	-0.072	0	-0.258	0.120	No stable signal.
Total firearm deaths	1.341	4	1.129	1.592	Positive, trend-sensitive.

Arkansas recoding as 2021 or 2023 retained the sign of all five outcomes. Firearm suicide, total suicide, non-firearm suicide, and total firearm deaths retained $p < 0.05$ across both Arkansas codings; firearm homicide did not.

Supplementary Note 9. Heterogeneity and policy-feature checks

Rurality was the clearest recurring heterogeneity dimension: firearm suicide, total suicide, and total firearm deaths had positive high-rurality interactions. These models are exploratory because state-level subgroup indicators are coarse and adoption is politically patterned.

Outcome/dimension	Interaction	p
Firearm suicide, high rurality	0.870	0.003
Total suicide, high rurality	1.325	0.002
Total firearm deaths, high rurality	1.630	0.026
Non-firearm suicide, high gun ownership	0.549	0.001

Policy-feature heterogeneity did not identify a stable mechanism. None of the 15 training, carry-permit background-check, or misdemeanor-screen interaction rows had $p < 0.05$. Five rows were sparse or unavailable because the feature-specific clean-adopter comparison group was absent.

Supplementary Note 10. Homicide missingness audit

Firearm-homicide outcome rows have fewer observations than suicide and total-firearm rows because some state-year rates are unavailable or suppressed. The missingness audit shows that all missing firearm-homicide rows occur in the smallest-population state-years.

Table 3: **Firearm-homicide missingness audit.**

Group	Possible	Observed	Missing	Missing pct.
All state-years	1300	1199	101	7.8%
Adopter-state state-years	702	636	66	9.4%
Non-adopter-state state-years	598	563	35	5.9%
Q1 smallest	325	224	101	31.1%
Q2	325	325	0	0.0%
Q3	325	325	0	0.0%
Q4 largest	325	325	0	0.0%

Missing firearm-homicide rates are concentrated in the smallest-population state-years. No state-year in population quartiles 2–4 is missing the firearm-homicide rate.

State	Missing	Possible	Missing pct.
Vermont	21	26	80.8%
New Hampshire	16	26	61.5%
North Dakota	15	26	57.7%
South Dakota	14	26	53.8%
Wyoming	14	26	53.8%
Hawaii	13	26	50.0%
Maine	6	26	23.1%
Delaware	1	26	3.8%
Montana	1	26	3.8%

Supplementary Note 11. Full diagnostic source files

The following repository outputs provide the full machine-readable diagnostic trail used to write the manuscript and this supplement:

- `outputs/tables/did/twfe.did.main.results.csv`
- `outputs/tables/did/twfe.did.firearm.law.control.summary.csv`
- `outputs/tables/did/twfe.did.nonfirearm.confounder.summary.csv`
- `outputs/tables/did/twfe.did.phase3b2.confounder.summary.csv`
- `outputs/tables/main/welch.change.score.results.csv`
- `outputs/tables/modern.did/modern.did.summary.csv`
- `outputs/tables/modern.did/event.time.never.control.estimates.csv`
- `outputs/tables/robustness/robustness.summary.csv`
- `outputs/tables/robustness/arkansas.treatment.sensitivity.summary.csv`
- `outputs/tables/heterogeneity/heterogeneity.did.results.csv`
- `outputs/tables/mechanism/mechanism.heterogeneity.results.csv`
- `data/policy/permitless.carry.legal.audit.csv`